

KArray: A public archive for microarray-based functional genomics data

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Introduction

In recent years, large-scale genomic data have become the cornerstone of modern biomedical and life science research. The increasing availability of multi-omics data has enabled data-driven research to unravel complex biological mechanisms and support biomarker discovery. Over the past few decades, microarray technology has played a pivotal role in advancing functional genomics by enabling large-scale gene expression profiling and molecular characterization of diverse biological conditions. However, a substantial number of datasets remain scattered across individual laboratories or unpublished, limiting their accessibility. Therefore, establishing a well-curated public repository is essential for facilitating underutilized data and supporting the growing demand for multi-omics research. To address this need, we developed KArray (<https://kbds.re.kr/karray>), a public microarray data repository designed to collect, curate, and provide access to microarray datasets. By offering standardized metadata annotation as well as user-friendly search, filtering, and download functionalities, KArray will enhance the availability of functional genomics resources.

Graphical abstract

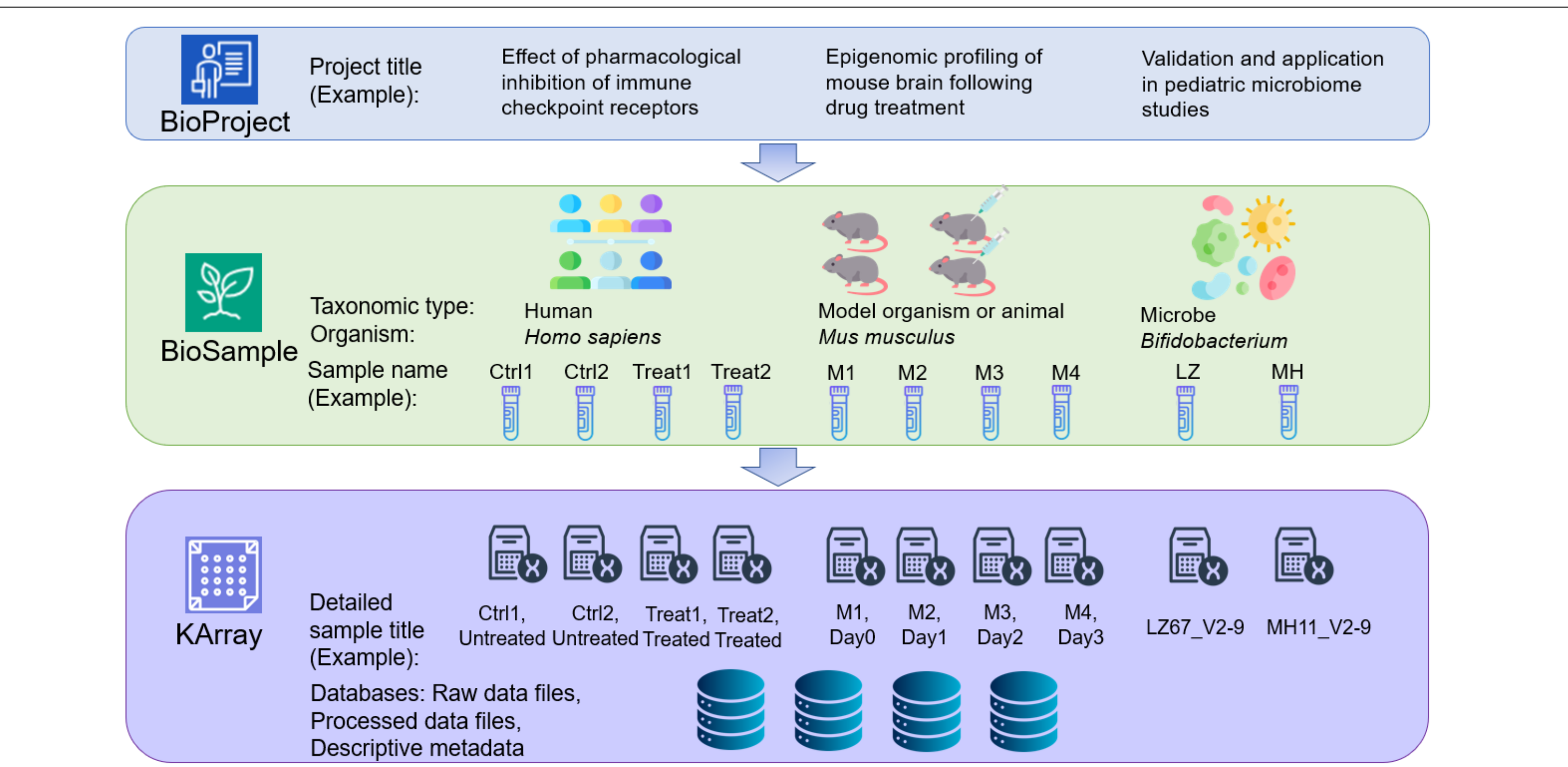


Figure 1. Schematic illustration of KArray data submission and structure

KArray integrates information from BioProject, BioSample, and KArray dataset submissions. Data Submission begins with BioProject, which defines the overall study, and BioSample, which specifies organisms and includes sample attributes. Within KArray, data are expanded into detailed sample titles that capture such information as treatment conditions, time points, and microbial species. Each entry links to raw data files, processed data files, and metadata.

Main

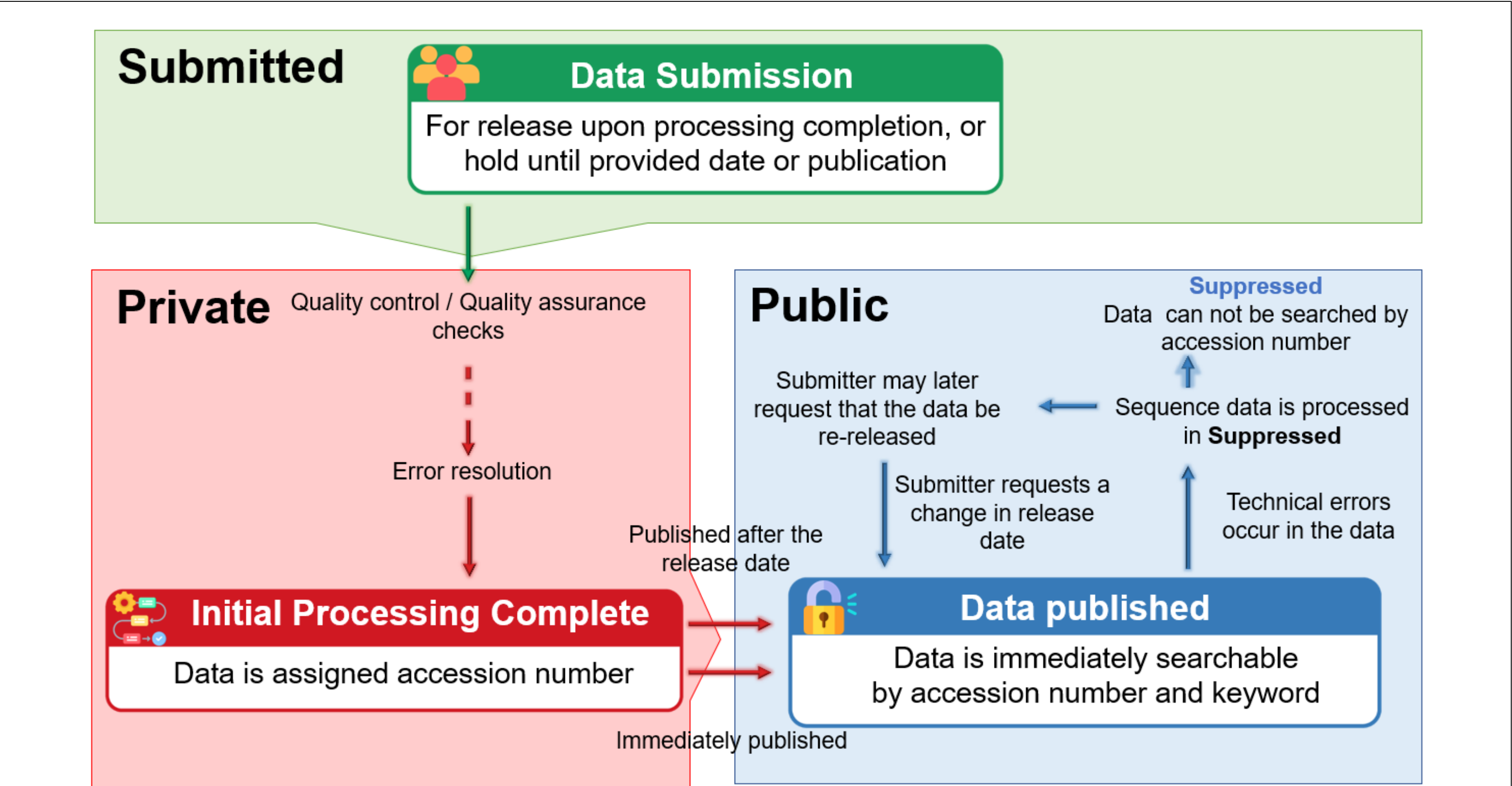


Figure 2. Workflow of data submission and curation in KArray

Submitted datasets undergo an internal quality control, quality assurance, and error resolution process. After processing, data are released publicly or placed in a suppressed state if technical errors occur.

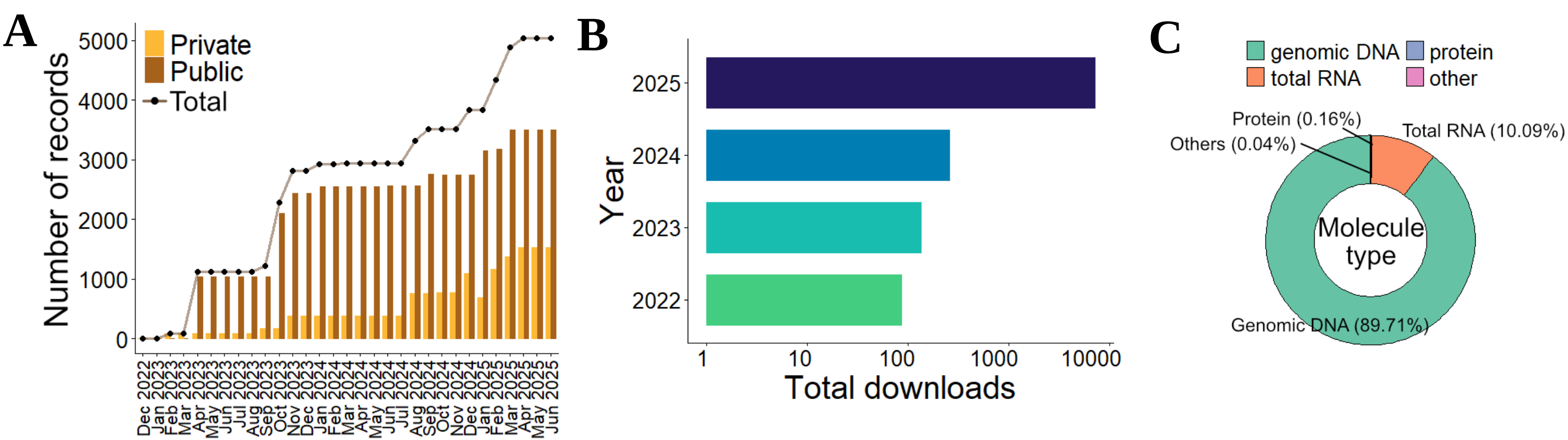


Figure 3. Statistical summary of KArray

(A) Number of KArray data. Each bar is divided into Public and Private records, while the dotted line indicates the cumulative total number of records across months. (B) Annual download counts of registered data. Each bar represents the total number of downloads per year. (C) Distribution of data by molecule type. The donut chart shows the proportion of genomic DNA, total RNA, protein, and other molecule types, with genomic DNA accounting for the largest share.

Main

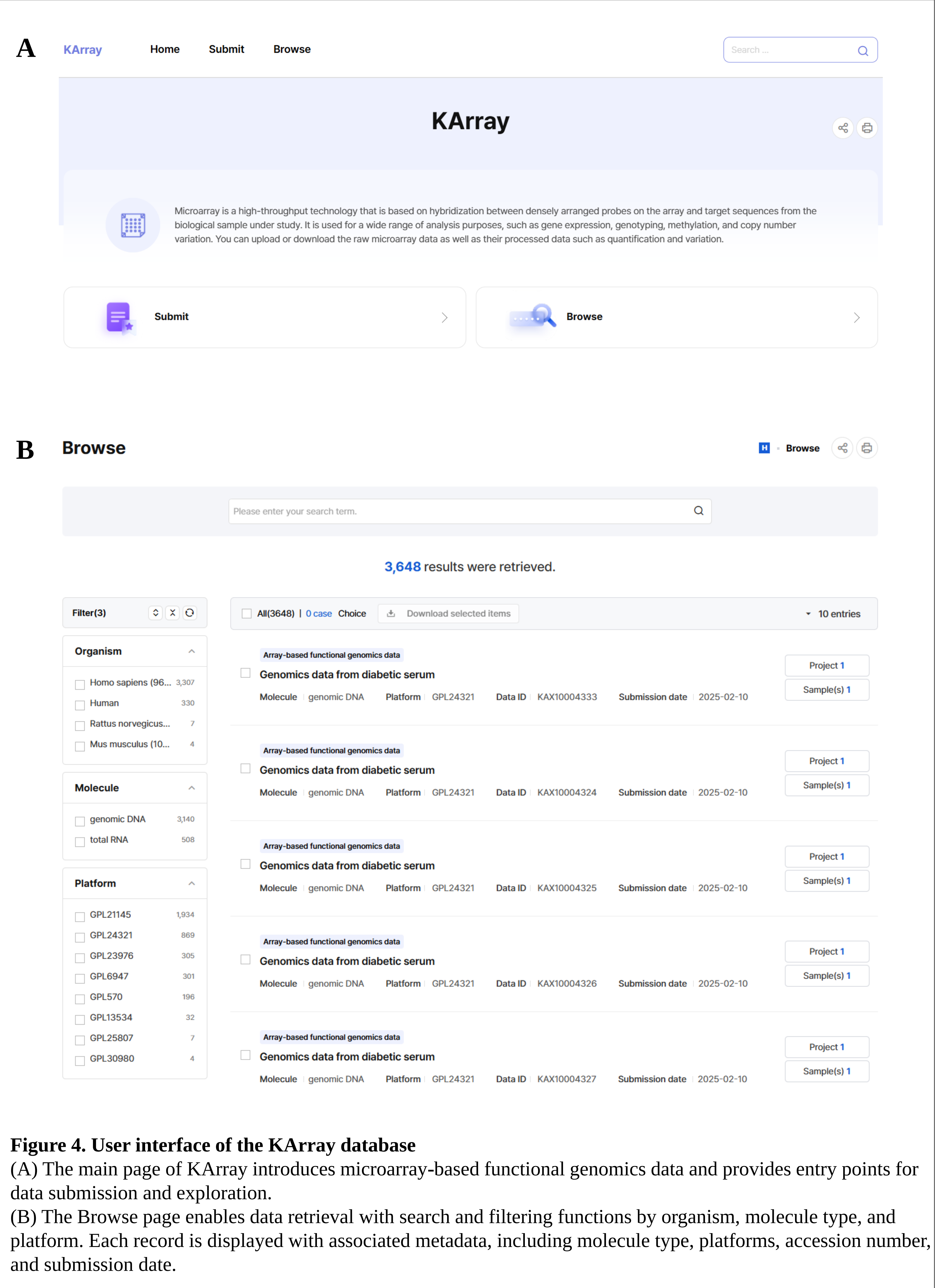


Figure 4. User interface of the KArray database

(A) The main page of KArray introduces microarray-based functional genomics data and provides entry points for data submission and exploration. (B) The Browse page enables data retrieval with search and filtering functions by organism, molecule type, and platform. Each record is displayed with associated metadata, including molecule type, platforms, accession number, and submission date.

Conclusions and future directions

KArray addresses the need for a centralized and well-curated repository for microarray-derived data. KArray ensures consistent data quality and interoperability with global resources by adopting metadata standards aligned with the Minimum Information About a Microarray Experiment (MIAME) guidelines, an internationally standardized format for microarray data. As of August 2025, KArray stores data related to over 3,000 samples, encompassing a variety of experimental methods and biological samples. KArray offers category filtering and user-friendly data retrieval functions, supporting data-centric approaches in functional genomics research. Moving forward, we plan to broaden KArray's scope by incorporating processed data files from next-generation sequencing (NGS) technologies, including RNA sequencing (RNA-seq), chromatin immunoprecipitation sequencing (ChIP-seq), single-cell RNA sequencing (scRNA-seq), single-cell assay for transposase-accessible chromatin with sequencing (scATAC-seq), and spatial transcriptomics. Additionally, interactive analysis and visualization tools will be developed to enable researchers to explore gene expression patterns and analytical results. Through continuous development, KArray aims to evolve into a comprehensive functional genomics data platform, facilitating data reuse for advancing genomic and transcriptomic studies.

References

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